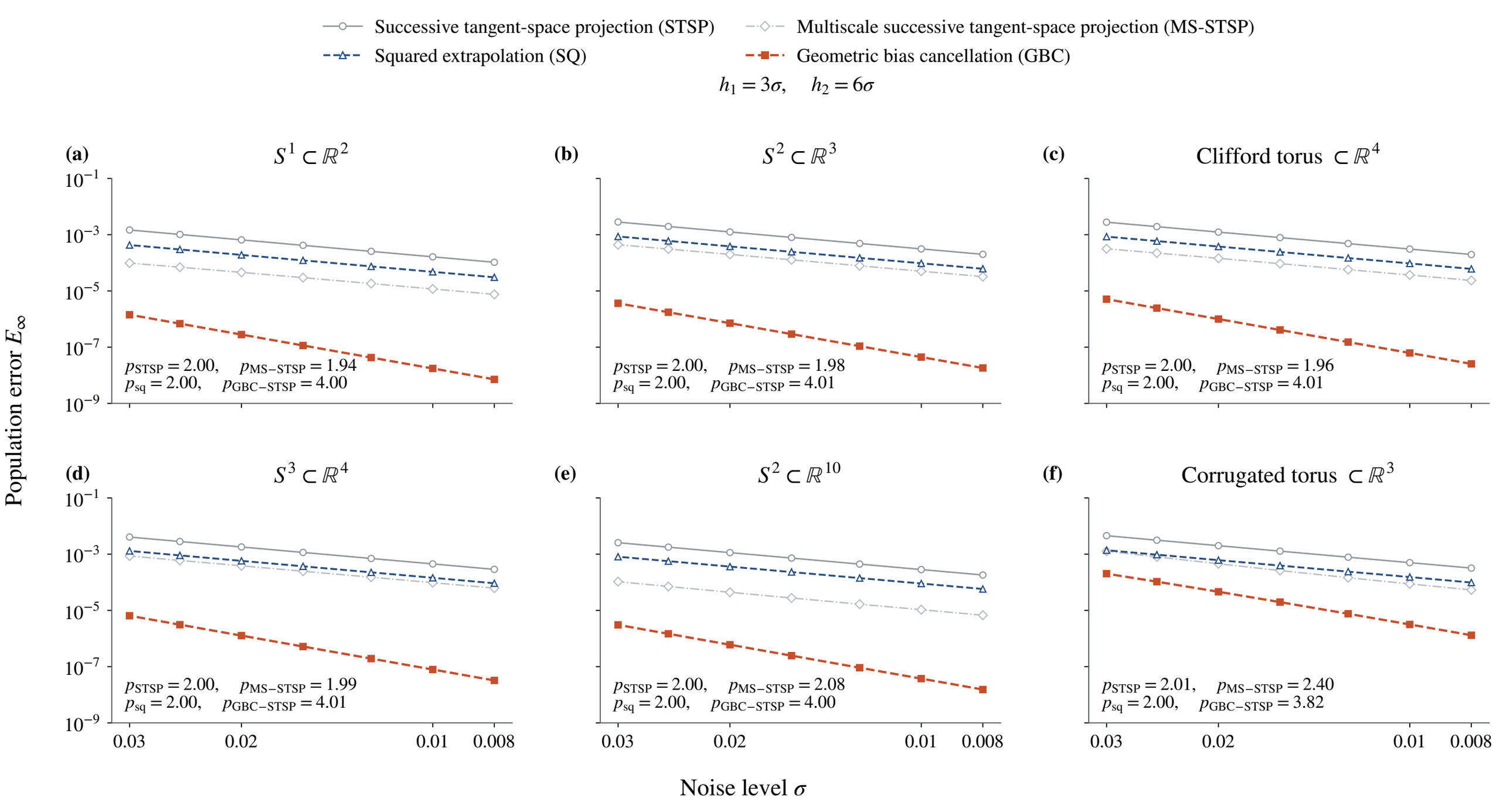
Successive tangent-space projection (STSP)
Multiscale successive tangent-space projection (MS-STSP)
Squared extrapolation (SQ)
Geometric bias cancellation (GBC)
$h_1 = 3\sigma, \quad h_2 = 6\sigma$
(a)
$S^1 \subset \mathbb{R}^2$
$p_{\mathrm{STSP}} = 2.00, \quad p_{\mathrm{MS-STSP}} = 1.94$
$p_{\mathrm{sq}} = 2.00, \quad p_{\mathrm{GBC-STSP}} = 4.00$
(b)
$S^2 \subset \mathbb{R}^3$
$p_{\mathrm{STSP}} = 2.00, \quad p_{\mathrm{MS-STSP}} = 1.98$
$p_{\mathrm{sq}} = 2.00, \quad p_{\mathrm{GBC-STSP}} = 4.01$
(c)
Clifford torus $\subset \mathbb{R}^4$
$p_{\mathrm{STSP}} = 2.00, \quad p_{\mathrm{MS-STSP}} = 1.96$
$p_{\mathrm{sq}} = 2.00, \quad p_{\mathrm{GBC-STSP}} = 4.01$
(d)
$S^3 \subset \mathbb{R}^4$
$p_{\mathrm{STSP}} = 2.00, \quad p_{\mathrm{MS-STSP}} = 1.99$
$p_{\mathrm{sq}} = 2.00, \quad p_{\mathrm{GBC-STSP}} = 4.01$
(e)
$S^2 \subset \mathbb{R}^{10}$
$p_{\mathrm{STSP}} = 2.00, \quad p_{\mathrm{MS-STSP}} = 2.08$
$p_{\mathrm{sq}} = 2.00, \quad p_{\mathrm{GBC-STSP}} = 4.00$
(f)
Corrugated torus $\subset \mathbb{R}^3$
$p_{\mathrm{STSP}} = 2.01, \quad p_{\mathrm{MS-STSP}} = 2.40$
$p_{\mathrm{sq}} = 2.00, \quad p_{\mathrm{GBC-STSP}} = 3.82$
Population error E_∞
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
0.03
0.02
0.01
0.008
Noise level σ